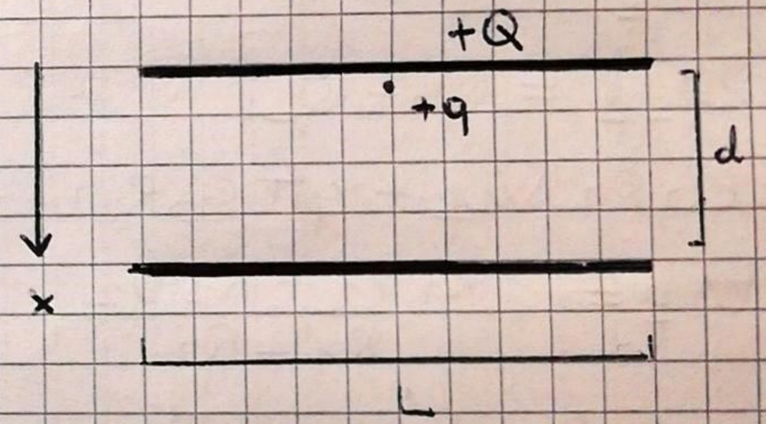
APPELLO 22/02/2016

ESERCIZIO 1



$$\underline{E} = \frac{\sigma}{\varepsilon_0}\,\underline{u}_x = \frac{Q}{L^2\varepsilon_0}\,\underline{u}_x$$

$$\Delta V = Ed = \frac{Qd}{L^2\varepsilon_0}$$

$$U = \frac{1}{2}\frac{Q^2}{C} = \frac{Q^2 d}{2\varepsilon_0 L^2}$$

$$L = q\Delta V = \Delta E_c = \frac{1}{2} m v_f^2$$

$$\Rightarrow v_f = \sqrt{2\frac{q\Delta V}{m}} = \sqrt{\frac{2qQd}{mL^2\varepsilon_0}}$$

ESERCIZIO 2

$[E_t] = E_{t2} - E_{t1} = 0$

$\Rightarrow E_{t1} = E_{t2}$

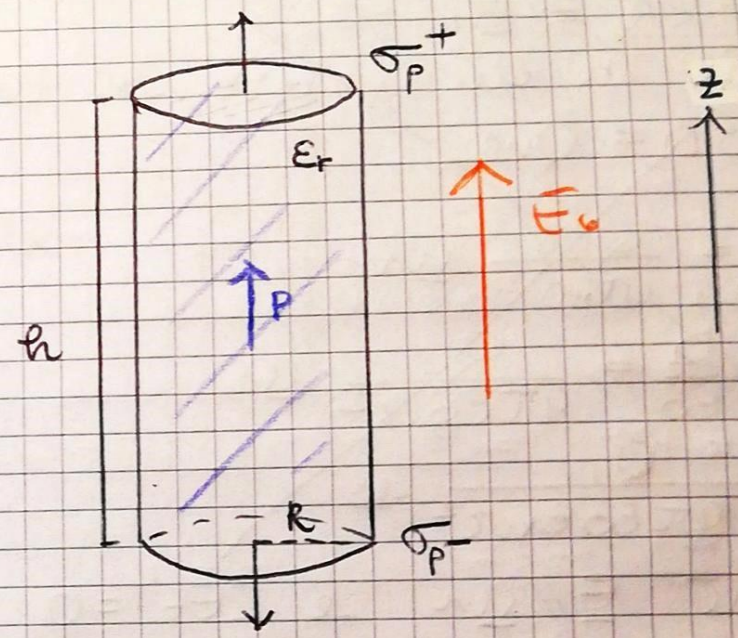
Ma $\underline{E} \parallel \underline{u}_t$

$\Rightarrow \underline{E}_{INT} = \underline{E}_0$

$\underline{P} = \chi \underline{E} = \varepsilon_0(\varepsilon_r - 1)\underline{E}_0$

$\sigma_p^+ = \varepsilon_0(\varepsilon_r - 1)E_0$

$\sigma_p^- = -\varepsilon_0(\varepsilon_r - 1)E_0$



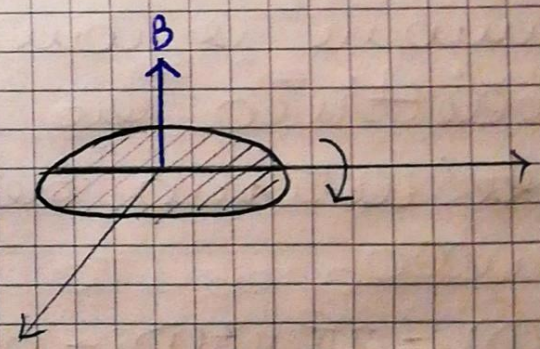
ESERCIZIO 3

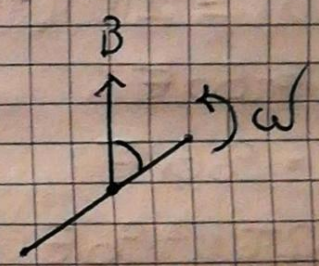
$\Phi(\underline{B}) = B\pi R^2 \cos(\omega t)$

$f = -\dfrac{d\Phi}{dt} = \omega B \pi R^2 \operatorname{sen}(\omega t)$

$P_{FOR} = P_J = \dfrac{f^2}{R} = \dfrac{\omega^2 B^2 \pi^2 R^4 \sin^2(\omega t)}{R}$

$\Rightarrow \overline{P}_J = \dfrac{\omega^2 B^2 \pi^2 R^4}{2R}$





~~ESERCIZIO 4~~